**Annual Report - General Order No. R5-2007-0035**
*Reporting period 01/01/2023 to 12/31/2023.*

DAIRY FACILITY INFORMATION

**A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY:** Ed Machado Dairy

Physical address of dairy:

| 5001 Shiloh RD | Modesto | Stanislaus | 95358 |
|---|---|---|---|
| Number and Street | City | County | Zip Code |

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1985

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X017-X066-X007-XXXX X017-X066-X008-XXXX

**B. OPERATORS**

Machado, Edward Anthony

Operator name: Machado, Edward Anthony

Telephone no.: (209) 599-2565 (Landline) (Cellular)

| 11708 E Moncure RD | Ripon | CA | 95366 |
|---|---|---|---|
| Mailing Address Number and Street | City | State | Zip Code |

**This operator is responsible for paying permit fees.**

**C. OWNERS**

Machado, Ed Jr.

Legal owner name: Machado, Ed Jr.

Telephone no.: (209) 599-2565 (Landline) (209) 481-0482 (Cellular)

| 11708 E Moncure | Ripon | CA | 95366 |
|---|---|---|---|
| Mailing Address Number and Street | City | State | Zip Code |

AVAILABLE NUTRIENTS

## A. HERD INFORMATION

| | Milk Cows | Dry Cows | Bred Heifers (15-24 mo.) | Heifers (7-14 mo. to breeding) | Calves (4-6 mo.) | Calves (0-3 mo.) |
|---|---|---|---|---|---|---|
| Number open confinement | 0 | 0 | 0 | 0 | 0 | 350 |
| Number under roof | 320 | 0 | 0 | 0 | 0 | 0 |
| Maximum number | 320 | 0 | 0 | 0 | 0 | 350 |
| Average number | 320 | 0 | 0 | 0 | 0 | 350 |
| Avg live weight (lbs) | 1,400 | 0 | 0 | 0 | | |

Predominant milk cow breed: Holstein

Average milk production: 72 *pounds per cow per day*

## B. MANURE GENERATED

Total manure excreted by the herd: 9,497.67 *tons per reporting period*

Total nitrogen from manure: 126,188.69 *lbs per reporting period*

After ammonia losses (30% loss applied): 88,332.08 *lbs per reporting period*

Total phosphorus from manure: 19,614.20 *lbs per reporting period*

Total potassium from manure: 61,703.43 *lbs per reporting period*

Total salt from manure: 150,672.00 *lbs per reporting period*

## C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 18,360,000 *gallons*

Total nitrogen generated: 32,801.66 *lbs*

Total phosphorus generated: 9,316.18 *lbs*

Total potassium generated: 53,595.93 *lbs*

Total salt generated: 511,887.64 *lbs*

| | | |
|---|---|---|
| | 18,360,000 | *gallons applied* |
| + | 0 | *gallons exported* |
| - | 0 | *gallons imported* |
| = | 18,360,000 | *gallons generated* |

## D. FRESH WATER SOURCES

| Source Description | Type |
|---|---|
| River Water | Surface water |

## E. SUBSURFACE (TILE) DRAINAGE SOURCES

*No subsurface (tile) drainage sources entered.*

## F. NUTRIENT IMPORTS

*No dry manure nutrient imports entered.*

*No process wastewater nutrient imports entered.*

| Date | Material type / Description | Quantity | Reporting basis | Moisture (%) | N (%) | P (%) | K (%) | Salt (%) |
|---|---|---|---|---|---|---|---|---|
| 06/12/2023 | Solid commercial fertilizer<br>*4-0-0* | 8.87 *ton* | Dry-weight | 0.1 | 4.000000 | 0.000000 | 0.000000 | 0.000000 |

| Material type | Total N (lbs) | Total P (lbs) | Total K (lbs) | Total salt (lbs) |
|---|---|---|---|---|
| Commercial fertilizer / Other | 708.89 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 0.00 | 0.00 | 0.00 | 0.00 |
| Total imports for all materials | 708.89 | 0.00 | 0.00 | 0.00 |

## G. NUTRIENT EXPORTS

| Date | Material type | Quantity | Reporting basis | Moisture (%) | Density (lbs/cu ft) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|---|
| 10/05/2023 | Corral solids | 525.00 *ton* | Dry-weight | 10.3 | | 20,900.00 | 4,620.00 | 23,800.00 | | 41.00 |

*No liquid nutrient exports entered.*

| Material type | Total N (lbs) | Total P (lbs) | Total K (lbs) | Total salt (lbs) |
|---|---|---|---|---|
| Dry manure | 19,684.67 | 4,351.35 | 22,416.03 | 386,158.50 |
| Process wastewater | 0.00 | 0.00 | 0.00 | 0.00 |
| Total exports for all materials | 19,684.67 | 4,351.35 | 22,416.03 | 386,158.50 |

APPLICATION AREA

## A. LIST OF LAND APPLICATION AREAS

| Field name | Controlled acres | Cropable acres | Total harvests | Type of waste applied | Parcel number |
|---|---|---|---|---|---|
| 1 | 15 | 15 | 2 | process wastewater | X017-X066-X008-XXXX |
| 2 | 28 | 28 | 2 | process wastewater | X017-X066-X007-XXXX |
| 3 | 28 | 28 | 2 | process wastewater | X017-X066-X009-XXXX |
| Totals for areas that were used for application | 71 | 71 | 6 | | |
| Totals for areas that were not used for application | | | | | |
| Land application area totals | 71 | 71 | 6 | | |

## B. CROPS AND HARVESTS

1

Field name: 1

11/04/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 15 Plant date: 11/04/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 04/28/2023 | 277.50 *ton* | As-is | | 71.8 | 4,720.00 | 1,050.00 | 9,190.00 | | 11.80 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 17.00 | 170.00 | 27.20 | 141.10 | 0.00 |
| Total actual harvest content | 18.50 | 174.64 | 38.85 | 340.03 | 1,231.21 |

05/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 15 Plant date: 05/20/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 09/11/2023 | 429.00 *ton* | As-is | | 70.0 | 4,090.00 | 740.00 | 6,220.00 | | 13.00 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 28.00 | 224.00 | 42.00 | 184.80 | 0.00 |
| Total actual harvest content | 28.60 | 233.95 | 42.33 | 355.78 | 2,230.80 |

## 2

Field name: 2

### 11/04/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough  Acres planted: 28  Plant date: 11/04/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 04/28/2023 | 534.80 *ton* | As-is | | 68.2 | 4,040.00 | 1,100.00 | 9,320.00 | | 9.70 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 17.00 | 170.00 | 27.20 | 141.10 | 0.00 |
| Total actual harvest content | 19.10 | 154.33 | 42.02 | 356.02 | 1,178.32 |

### 05/20/2023: Corn, silage

Crop: Corn, silage  Acres planted: 28  Plant date: 05/20/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 09/11/2023 | 814.80 *ton* | As-is | | 70.0 | 4,230.00 | 740.00 | 6,940.00 | | 14.40 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 28.00 | 224.00 | 42.00 | 184.80 | 0.00 |
| Total actual harvest content | 29.10 | 246.19 | 43.07 | 403.91 | 2,514.24 |

## 3

Field name: 3

### 11/04/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough  Acres planted: 28  Plant date: 11/04/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 04/28/2023 | 0.01 *ton* | As-is | | 0.1 | 0.00 | 0.00 | 0.00 | | 0.00 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 17.00 | 170.00 | 27.20 | 141.10 | 0.00 |
| Total actual harvest content | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |

3

06/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 28 Plant date: 06/12/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 09/11/2023 | 646.80 *ton* | As-is | | 70.0 | 4,350.00 | 970.00 | 6,480.00 | | 13.40 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 28.00 | 224.00 | 42.00 | 184.80 | 0.00 |
| Total actual harvest content | 23.10 | 200.97 | 44.81 | 299.38 | 1,857.24 |

NUTRIENT BUDGET

## A. LAND APPLICATIONS

### 1 - 11/04/2022: Oats, silage-soft dough

Field name: 1

Crop: Oats, silage-soft dough  Plant date: 11/04/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 01/20/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 125.72 | 37.01 | 207.97 | 1,451.10 | 528,000.00 *gal* |
| Application event totals | | 125.72 | 37.01 | 207.97 | 1,451.10 | |

| | | | | |
|---|---|---|---|---|
| 03/17/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 108.58 | 31.96 | 179.61 | 1,253.22 | 456,000.00 *gal* |
| Application event totals | | 108.58 | 31.96 | 179.61 | 1,253.22 | |

### 1 - 05/20/2023: Corn, silage

Field name: 1

Crop: Corn, silage  Plant date: 05/20/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 05/10/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 125.72 | 37.01 | 207.97 | 1,451.10 | 528,000.00 *gal* |
| River Water | Surface water | 2.49 | 0.00 | 0.00 | 449.80 | 1,650,000.00 *gal* |
| Application event totals | | 128.21 | 37.01 | 207.97 | 1,900.89 | |

| | | | | |
|---|---|---|---|---|
| 05/20/2023 | Sidedress | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| 4-0-0 | Solid commercial fertilizer | 10.00 | 0.00 | 0.00 | 0.00 | |
| Application event totals | | 10.00 | 0.00 | 0.00 | 0.00 | |

1 - 05/20/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/20/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| River Water | Surface water | 2.94 | 0.00 | 0.00 | 531.58 | 1,950,000.00 *gal* |
| Application event totals | | 2.94 | 0.00 | 0.00 | 531.58 | |

| | | | | |
|---|---|---|---|---|
| 07/02/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 32.85 | 7.96 | 49.40 | 432.60 | 480,000.00 *gal* |
| River Water | Surface water | 2.26 | 0.00 | 0.00 | 408.91 | 1,500,000.00 *gal* |
| Application event totals | | 35.11 | 7.96 | 49.40 | 841.51 | |

| | | | | |
|---|---|---|---|---|
| 07/13/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 32.85 | 7.96 | 49.40 | 432.60 | 480,000.00 *gal* |
| River Water | Surface water | 2.26 | 0.00 | 0.00 | 408.91 | 1,500,000.00 *gal* |
| Application event totals | | 35.11 | 7.96 | 49.40 | 841.51 | |

| | | | | |
|---|---|---|---|---|
| 07/24/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| River Water | Surface water | 2.83 | 0.00 | 0.00 | 511.13 | 1,875,000.00 *gal* |
| Application event totals | | 2.83 | 0.00 | 0.00 | 511.13 | |

| | | | | |
|---|---|---|---|---|
| 08/04/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 31.20 | 7.56 | 46.93 | 410.97 | 456,000.00 *gal* |
| River Water | Surface water | 2.15 | 0.00 | 0.00 | 388.46 | 1,425,000.00 *gal* |
| Application event totals | | 33.35 | 7.56 | 46.93 | 799.43 | |

## 1 - 05/20/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/16/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 31.20 | 7.56 | 46.93 | 410.97 | 456,000.00 *gal* |
| River Water | Surface water | 2.15 | 0.00 | 0.00 | 388.46 | 1,425,000.00 *gal* |
| Application event totals | | 33.35 | 7.56 | 46.93 | 799.43 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/29/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 29.56 | 7.16 | 44.46 | 389.34 | 432,000.00 *gal* |
| River Water | Surface water | 2.04 | 0.00 | 0.00 | 368.01 | 1,350,000.00 *gal* |
| Application event totals | | 31.60 | 7.16 | 44.46 | 757.36 | |

## 2 - 11/04/2022: Oats, silage-soft dough

Field name: 2

Crop: Oats, silage-soft dough

Plant date: 11/04/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 12/16/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 38.34 | 13.53 | 73.53 | 1,785.35 | 960,000.00 *gal* |
| Application event totals | | 38.34 | 13.53 | 73.53 | 1,785.35 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 02/12/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 73.47 | 21.63 | 121.54 | 848.04 | 576,000.00 *gal* |
| Application event totals | | 73.47 | 21.63 | 121.54 | 848.04 | |

## 2 - 11/04/2022: Oats, silage-soft dough

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 03/25/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 97.97 | 28.84 | 162.06 | 1,130.72 | 768,000.00 *gal* |
| Application event totals | | 97.97 | 28.84 | 162.06 | 1,130.72 | |

## 2 - 05/20/2023: Corn, silage

Field name: 2

Crop: Corn, silage

Plant date: 05/20/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 05/10/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 125.52 | 36.95 | 207.63 | 1,448.74 | 984,000.00 *gal* |
| River Water | Surface water | 2.48 | 0.00 | 0.00 | 449.07 | 3,075,000.00 *gal* |
| Application event totals | | 128.00 | 36.95 | 207.63 | 1,897.80 | |

| | | | | |
|---|---|---|---|---|
| 05/20/2023 | Sidedress | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| 4-0-0 | Solid commercial fertilizer | 10.00 | 0.00 | 0.00 | 0.00 | |
| Application event totals | | 10.00 | 0.00 | 0.00 | 0.00 | |

| | | | | |
|---|---|---|---|---|
| 06/20/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| River Water | Surface water | 2.97 | 0.00 | 0.00 | 536.69 | 3,675,000.00 *gal* |
| Application event totals | | 2.97 | 0.00 | 0.00 | 536.69 | |

## 2 - 05/20/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/02/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 32.55 | 7.89 | 48.96 | 428.74 | 888,000.00 *gal* |
| River Water | Surface water | 2.24 | 0.00 | 0.00 | 405.25 | 2,775,000.00 *gal* |
| Application event totals | | 34.79 | 7.89 | 48.96 | 834.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/13/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 32.55 | 7.89 | 48.96 | 428.74 | 888,000.00 *gal* |
| River Water | Surface water | 2.24 | 0.00 | 0.00 | 405.25 | 2,775,000.00 *gal* |
| Application event totals | | 34.79 | 7.89 | 48.96 | 834.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/24/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| River Water | Surface water | 2.91 | 0.00 | 0.00 | 525.74 | 3,600,000.00 *gal* |
| Application event totals | | 2.91 | 0.00 | 0.00 | 525.74 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/04/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 31.67 | 7.67 | 47.64 | 417.15 | 864,000.00 *gal* |
| River Water | Surface water | 2.18 | 0.00 | 0.00 | 394.30 | 2,700,000.00 *gal* |
| Application event totals | | 33.85 | 7.67 | 47.64 | 811.46 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/16/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 31.67 | 7.67 | 47.64 | 417.15 | 864,000.00 *gal* |
| River Water | Surface water | 2.18 | 0.00 | 0.00 | 394.30 | 2,700,000.00 *gal* |
| Application event totals | | 33.85 | 7.67 | 47.64 | 811.46 | |

## 2 - 05/20/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/29/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 29.91 | 7.25 | 44.99 | 393.98 | 816,000.00 *gal* |
| River Water | Surface water | 2.06 | 0.00 | 0.00 | 372.40 | 2,550,000.00 *gal* |
| Application event totals | | 31.97 | 7.25 | 44.99 | 766.37 | |

## 3 - 11/04/2022: Oats, silage-soft dough

Field name: 3

Crop: Oats, silage-soft dough

Plant date: 11/04/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 10/06/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| River Water | Surface water | 8.65 | 0.00 | 0.00 | 616.93 | 3,000,000.00 *gal* |
| Application event totals | | 8.65 | 0.00 | 0.00 | 616.93 | |

| | | | | |
|---|---|---|---|---|
| 10/06/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 38.34 | 13.53 | 73.53 | 1,785.35 | 960,000.00 *gal* |
| Application event totals | | 38.34 | 13.53 | 73.53 | 1,785.35 | |

| | | | | |
|---|---|---|---|---|
| 11/29/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 42.17 | 14.89 | 80.88 | 1,963.89 | 1,056,000.00 *gal* |
| Application event totals | | 42.17 | 14.89 | 80.88 | 1,963.89 | |

| | | | | |
|---|---|---|---|---|
| 02/11/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 73.47 | 21.63 | 121.54 | 848.04 | 576,000.00 *gal* |
| Application event totals | | 73.47 | 21.63 | 121.54 | 848.04 | |

## 3 - 06/12/2023: Corn, silage

Field name: 3

Crop: Corn, silage

Plant date: 06/12/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/01/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 122.46 | 36.05 | 202.57 | 1,413.40 | 960,000.00 *gal* |
| River Water | Surface water | 2.42 | 0.00 | 0.00 | 438.11 | 3,000,000.00 *gal* |
| Application event totals | | 124.88 | 36.05 | 202.57 | 1,851.52 | |

| | | | | |
|---|---|---|---|---|
| 06/12/2023 | Sidedress | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| 4-0-0 | Solid commercial fertilizer | 10.00 | 0.00 | 0.00 | 0.00 | |
| Application event totals | | 10.00 | 0.00 | 0.00 | 0.00 | |

| | | | | |
|---|---|---|---|---|
| 07/12/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| River Water | Surface water | 2.91 | 0.00 | 0.00 | 525.74 | 3,600,000.00 *gal* |
| Application event totals | | 2.91 | 0.00 | 0.00 | 525.74 | |

| | | | | |
|---|---|---|---|---|
| 07/23/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 31.67 | 7.67 | 47.64 | 417.15 | 864,000.00 *gal* |
| River Water | Surface water | 2.91 | 0.00 | 0.00 | 525.74 | 3,600,000.00 *gal* |
| Application event totals | | 34.58 | 7.67 | 47.64 | 942.89 | |

| | | | | |
|---|---|---|---|---|
| 08/03/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 31.67 | 7.67 | 47.64 | 417.15 | 864,000.00 *gal* |
| River Water | Surface water | 2.18 | 0.00 | 0.00 | 394.30 | 2,700,000.00 *gal* |
| Application event totals | | 33.85 | 7.67 | 47.64 | 811.46 | |

3 - 06/12/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/15/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 30.79 | 7.46 | 46.31 | 405.57 | 840,000.00 *gal* |
| River Water | Surface water | 2.12 | 0.00 | 0.00 | 383.35 | 2,625,000.00 *gal* |
| Application event totals | | 32.91 | 7.46 | 46.31 | 788.92 | |

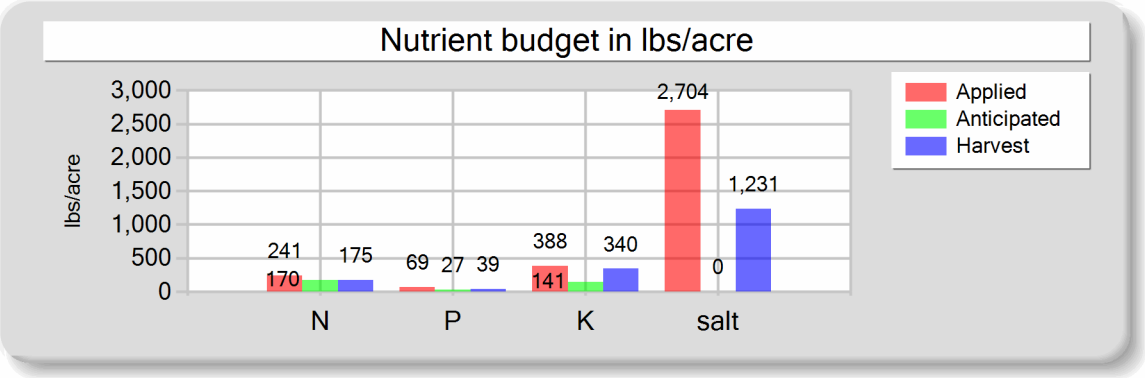
| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/28/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 29.91 | 7.25 | 44.99 | 393.98 | 816,000.00 *gal* |
| River Water | Surface water | 2.06 | 0.00 | 0.00 | 372.40 | 2,550,000.00 *gal* |
| Application event totals | | 31.97 | 7.25 | 44.99 | 766.37 | |

## B. NUTRIENT BUDGET

1 - 11/04/2022: Oats, silage-soft dough

Field name: 1 Crop: Oats, silage-soft dough Plant date: 11/04/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 234.30 | 68.98 | 387.58 | 2,704.31 |
| Fresh water | 0.00 | 0.00 | 0.00 | 0.00 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 241.30 | 68.98 | 387.58 | 2,704.31 |
| Anticipated crop nutrient removal | 170.00 | 27.20 | 141.10 | 0.00 |
| Actual crop nutrient removal | 174.64 | 38.85 | 340.03 | 1,231.21 |
| Nutrient balance | 66.66 | 30.13 | 47.55 | 1,473.10 |
| Applied to removed ratio | 1.38 | 1.78 | 1.14 | 2.20 |

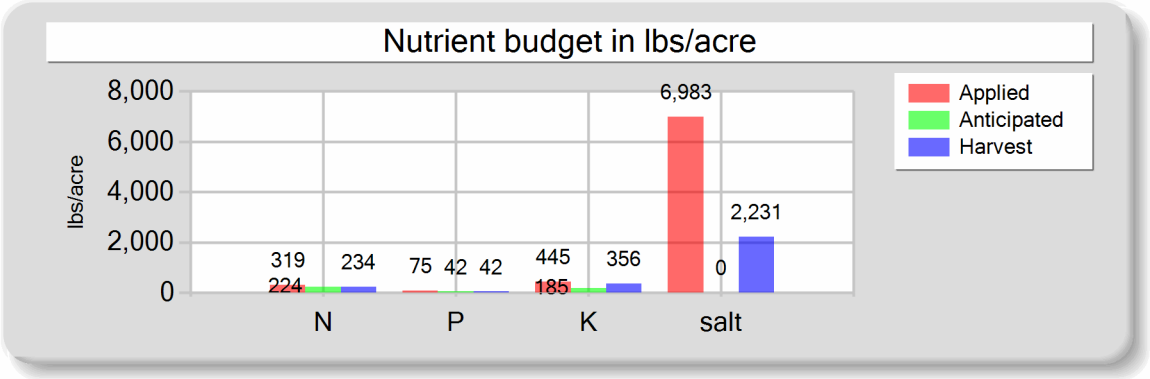
| Fresh water applied | |
|---|---|
| 0.00 | *gallons* |
| 0.00 | *acre-inches* |
| 0.00 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 984,000.00 | *gallons* |
| 36.24 | *acre-inches* |
| 2.42 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## 1 - 05/20/2023: Corn, silage

Field name: 1 Crop: Corn, silage Plant date: 05/20/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 10.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 283.38 | 75.21 | 445.10 | 3,527.60 |
| Fresh water | 19.11 | 0.00 | 0.00 | 3,455.25 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 319.49 | 75.21 | 445.10 | 6,982.85 |
| Anticipated crop nutrient removal | 224.00 | 42.00 | 184.80 | 0.00 |
| Actual crop nutrient removal | 233.95 | 42.33 | 355.78 | 2,230.80 |
| Nutrient balance | 85.54 | 32.88 | 89.32 | 4,752.05 |
| Applied to removed ratio | 1.37 | 1.78 | 1.25 | 3.13 |

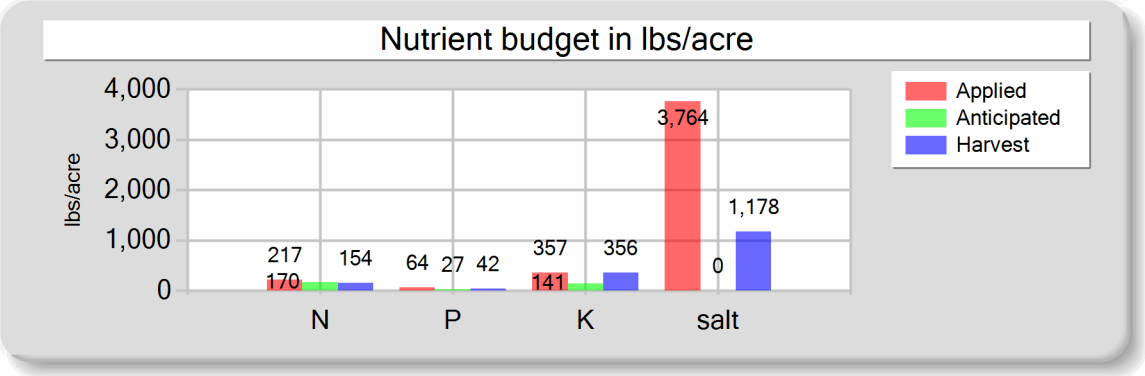
| Fresh water applied | |
|---|---|
| 12,675,000.00 | *gallons* |
| 466.78 | *acre-inches* |
| 31.12 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 2,832,000.00 | *gallons* |
| 104.29 | *acre-inches* |
| 6.95 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## 2 - 11/04/2022: Oats, silage-soft dough

Field name: 2 Crop: Oats, silage-soft dough Plant date: 11/04/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 209.78 | 64.00 | 357.13 | 3,764.12 |
| Fresh water | 0.00 | 0.00 | 0.00 | 0.00 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 216.78 | 64.00 | 357.13 | 3,764.12 |
| Anticipated crop nutrient removal | 170.00 | 27.20 | 141.10 | 0.00 |
| Actual crop nutrient removal | 154.33 | 42.02 | 356.02 | 1,178.32 |
| Nutrient balance | 62.45 | 21.98 | 1.10 | 2,585.80 |
| Applied to removed ratio | 1.40 | 1.52 | 1.00 | 3.19 |

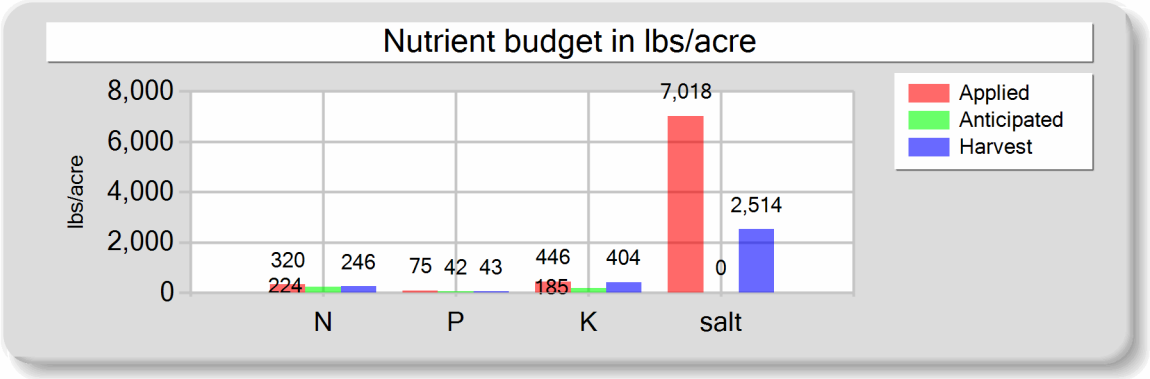
| Fresh water applied |
|---|
| 0.00 *gallons* |
| 0.00 *acre-inches* |
| 0.00 *inches/acre* |

| Process wastewater applied |
|---|
| 2,304,000.00 *gallons* |
| 84.85 *acre-inches* |
| 3.03 *inches/acre* |

| Total harvests for the crop |
|---|
| 1 *harvests* |

## 2 - 05/20/2023: Corn, silage

Field name: 2 Crop: Corn, silage Plant date: 05/20/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 10.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 283.88 | 75.32 | 445.82 | 3,534.51 |
| Fresh water | 19.26 | 0.00 | 0.00 | 3,482.99 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 320.15 | 75.32 | 445.82 | 7,017.51 |
| Anticipated crop nutrient removal | 224.00 | 42.00 | 184.80 | 0.00 |
| Actual crop nutrient removal | 246.19 | 43.07 | 403.91 | 2,514.24 |
| Nutrient balance | 73.96 | 32.25 | 41.92 | 4,503.27 |
| Applied to removed ratio | 1.30 | 1.75 | 1.10 | 2.79 |

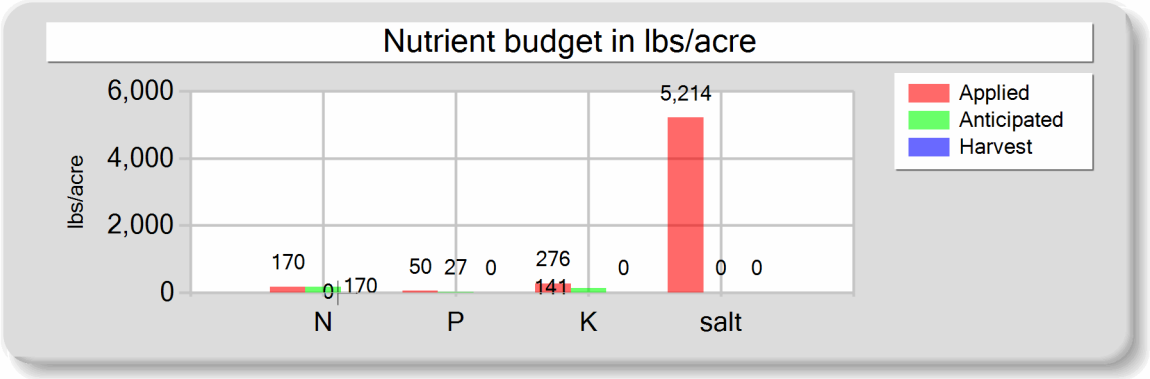
| Fresh water applied | |
|---|---|
| 23,850,000.00 | *gallons* |
| 878.31 | *acre-inches* |
| 31.37 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 5,304,000.00 | *gallons* |
| 195.33 | *acre-inches* |
| 6.98 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## 3 - 11/04/2022: Oats, silage-soft dough

Field name: 3 Crop: Oats, silage-soft dough Plant date: 11/04/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 153.99 | 50.05 | 275.96 | 4,597.28 |
| Fresh water | 8.65 | 0.00 | 0.00 | 616.93 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 169.63 | 50.05 | 275.96 | 5,214.22 |
| Anticipated crop nutrient removal | 170.00 | 27.20 | 141.10 | 0.00 |
| Actual crop nutrient removal | 0.00 | 0.00 | 0.00 | 0.00 |
| Nutrient balance | 169.63 | 50.05 | 275.96 | 5,214.22 |
| Applied to removed ratio | 0.00 | 0.00 | 0.00 | 0.00 |

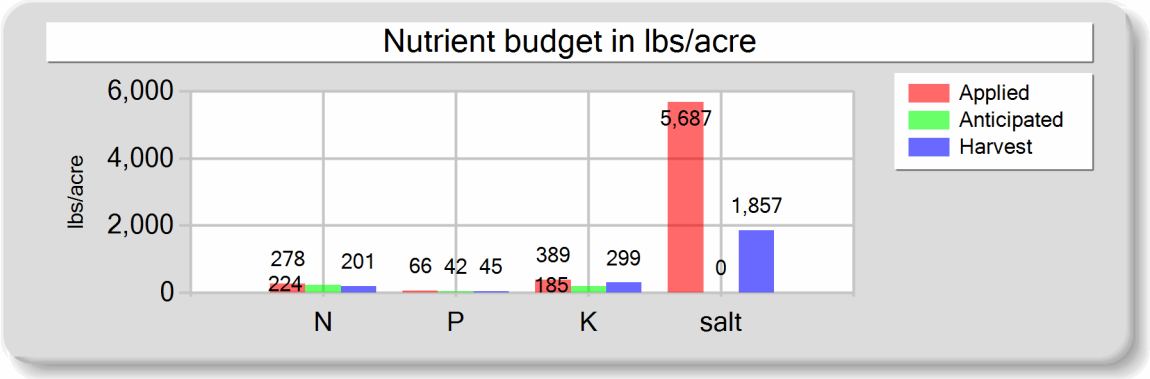
| Fresh water applied | |
|---|---|
| 3,000,000.00 | *gallons* |
| 110.48 | *acre-inches* |
| 3.95 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 2,592,000.00 | *gallons* |
| 95.45 | *acre-inches* |
| 3.41 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## 3 - 06/12/2023: Corn, silage

Field name: 3 Crop: Corn, silage Plant date: 06/12/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 10.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 246.51 | 66.11 | 389.15 | 3,047.26 |
| Fresh water | 14.60 | 0.00 | 0.00 | 2,639.63 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 278.11 | 66.11 | 389.15 | 5,686.89 |
| Anticipated crop nutrient removal | 224.00 | 42.00 | 184.80 | 0.00 |
| Actual crop nutrient removal | 200.97 | 44.81 | 299.38 | 1,857.24 |
| Nutrient balance | 77.14 | 21.29 | 89.78 | 3,829.65 |
| Applied to removed ratio | 1.38 | 1.48 | 1.30 | 3.06 |

| Fresh water applied | |
|---|---|
| 18,075,000.00 | *gallons* |
| 665.64 | *acre-inches* |
| 23.77 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 4,344,000.00 | *gallons* |
| 159.97 | *acre-inches* |
| 5.71 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

NUTRIENT ANALYSES

## A. MANURE ANALYSES

### Corral Solids

Sample and source description: Corral Solids

Sample date: 04/21/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 26.6 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Calcium (mg/kg) | Magnesium (mg/kg) | Sodium (mg/kg) | Sulfur (mg/kg) | Chloride (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 17,500.00 | 6,800.00 | 51,400.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | | 49.70 |
| **DL** | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 50.00 | | 0.01 |

### Corral Manure

Sample and source description: Corral Manure

Sample date: 09/12/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 10.3 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Calcium (mg/kg) | Magnesium (mg/kg) | Sodium (mg/kg) | Sulfur (mg/kg) | Chloride (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 20,900.00 | 4,620.00 | 23,800.00 | 17,100.00 | 6,210.00 | 4,030.00 | 8,830.00 | 1,210.00 | | 41.00 |
| **DL** | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 50.00 | | 0.01 |

## B. PROCESS WASTEWATER ANALYSES

### WWQ4

Sample and source description: WWQ4

Sample date: 10/20/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 134.00 | 86.80 | 0.00 | 0.00 | 47.30 | 257.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 3,960.00 | 6,240 |
| **DL** | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 2.00 | 1.00 | 0.10 | 1.00 | 10 |

## WWQ1

Sample and source description: WWQ1

Sample date: 02/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 428.00 | 241.00 | 0.00 | 0.00 | 126.00 | 708.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 6,370.00 | 4,940 |
| **DL** | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 2.00 | 1.00 | 0.10 | 1.00 | 10 |

## WWQ2

Sample and source description: WWQ2

Sample date: 04/21/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 792.00 | 339.00 | 0.00 | 0.00 | 113.00 | 1,450.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 8,410.00 | 6,280 |
| **DL** | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 2.00 | 1.00 | 0.10 | 1.00 | 10 |

## WWQ3

Sample and source description: WWQ3

Sample date: 07/20/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 123.00 | 67.20 | 0.00 | 0.00 | 29.80 | 185.00 | 67.50 | 51.30 | 98.10 | 900.00 | 0.00 | 91.90 | 149.00 | 2,310.00 | 1,620 |
| **DL** | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 2.00 | 1.00 | 0.10 | 1.00 | 10 |

## WWQ4

Sample and source description: WWQ4

Sample date: 11/29/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 473.00 | 302.00 | 0.00 | 0.00 | 70.10 | 807.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 7,050.00 | 6,920 |
| **DL** | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 2.00 | 1.00 | 0.10 | 1.00 | 10 |

## C. FRESH WATER ANALYSES

River Water

San Joaquin River

Sample description: San Joaquin River

Sample date: 06/06/2022 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 9.67 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 1,300.00 | 690 |
| **DL** | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 2.00 | 1.00 | 0.10 | 1.00 | 10 |

San Joaquin River

Sample description: San Joaquin River

Sample date: 11/14/2023 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 2.71 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 1,040.00 | 490 |
| **DL** | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 2.00 | 1.00 | 0.10 | 1.00 | 10 |

## D. SOIL ANALYSES

1

Soil

Sample and source description: Soil

Sample date: 12/01/2023 Source of analysis: Lab analysis

| | Nitrate-N (mg/kg) | Total P (mg/kg) | Soluble P (mg/kg) | K (mg/kg) | EC (µmhos/cm) | Organic matter (%) | Total salt (mg/kg) |
|---|---|---|---|---|---|---|---|
| **Value** | | | 52.00 | | | | |
| **DL** | | | 4.00 | | | | |

2

2

Soil

Sample and source description: Soil

Sample date: 12/01/2023 Source of analysis: Lab analysis

| | Nitrate-N (mg/kg) | Total P (mg/kg) | Soluble P (mg/kg) | K (mg/kg) | EC (µmhos/cm) | Organic matter (%) | Total salt (mg/kg) |
|---|---|---|---|---|---|---|---|
| **Value** | | | 54.00 | | | | |
| **DL** | | | 4.00 | | | | |

3

Soil

Sample and source description: Soil

Sample date: 12/01/2023 Source of analysis: Lab analysis

| | Nitrate-N (mg/kg) | Total P (mg/kg) | Soluble P (mg/kg) | K (mg/kg) | EC (µmhos/cm) | Organic matter (%) | Total salt (mg/kg) |
|---|---|---|---|---|---|---|---|
| **Value** | | | 41.00 | | | | |
| **DL** | | | 4.00 | | | | |

## E. PLANT TISSUE ANALYSES

1 - 11/04/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.8 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 4,720.00 | 1,050.00 | 9,190.00 | | 11.80 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

1 - 05/20/2023: Corn, silage

## 1 - 05/20/2023: Corn, silage

### Corn Silage

Sample and source description: Corn Silage

Sample date: 09/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 4,090.00 | 740.00 | 6,220.00 | | 13.00 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

## 2 - 11/04/2022: Oats, silage-soft dough

### Forage

Sample and source description: Forage

Sample date: 04/25/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.2 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 4,040.00 | 1,100.00 | 9,320.00 | | 9.70 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

## 2 - 05/20/2023: Corn, silage

### Corn Silage

Sample and source description: Corn Silage

Sample date: 09/11/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 4,230.00 | 740.00 | 6,940.00 | | 14.40 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

3 - 11/04/2022: Oats, silage-soft dough

Forage - Fallow Flooded

Sample and source description: Forage - Fallow Flooded

Sample date: 11/04/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 0.1 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 0.00 | 0.00 | 0.00 | | 0.00 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

3 - 06/12/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/11/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 4,350.00 | 970.00 | 6,480.00 | | 13.40 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

## F. SUBSURFACE (TILE) DRAINAGE ANALYSES

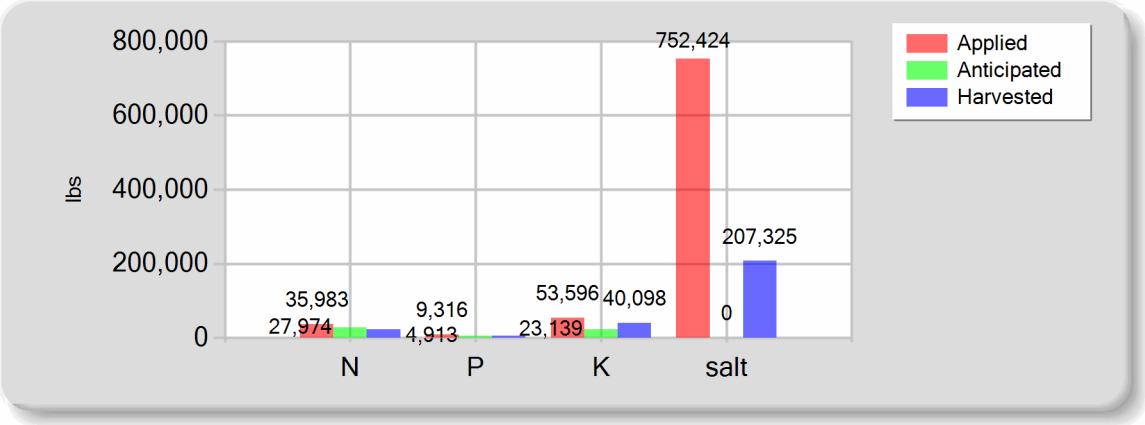
*No subsurface (tile) drainage analyses entered.*

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

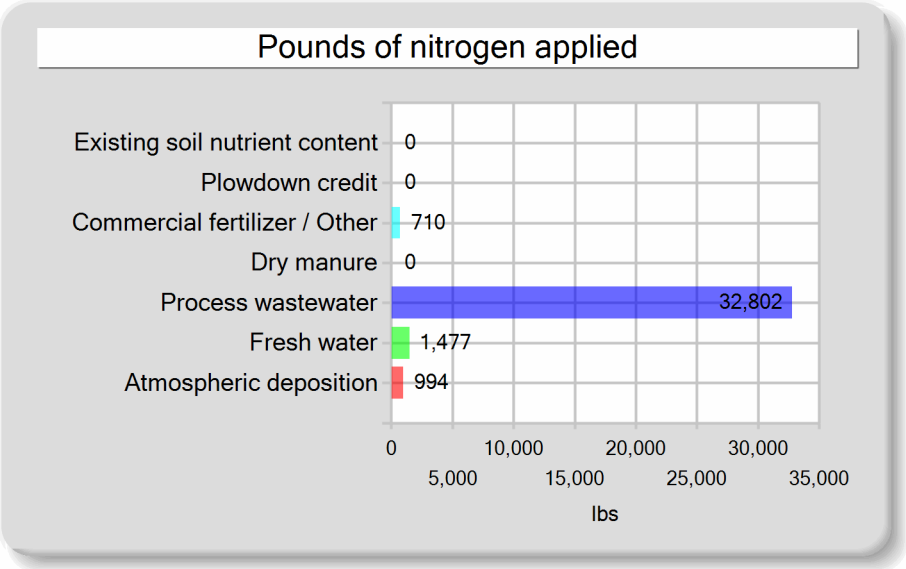
## A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

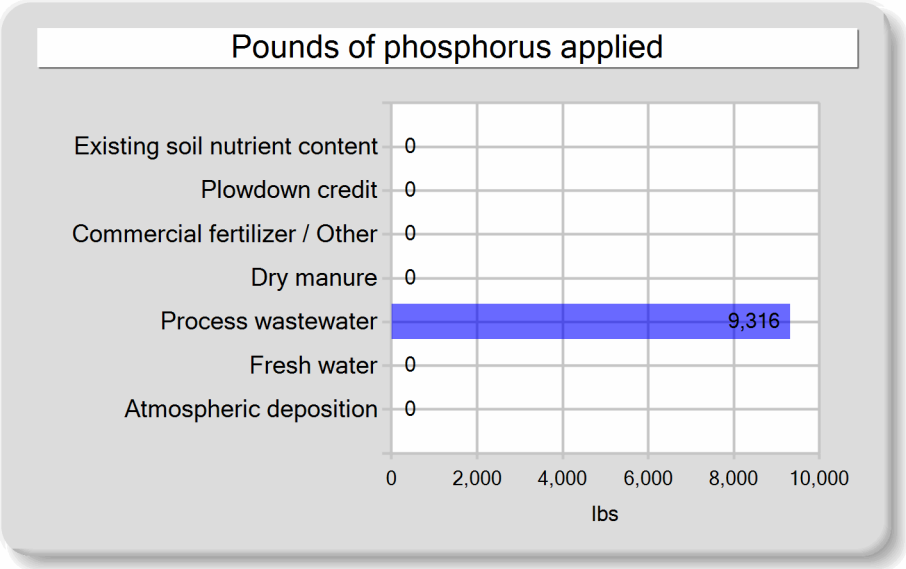
| | Total N (lbs) | Total P (lbs) | Total K (lbs) | Total salt (lbs) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 710.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 32,801.66 | 9,316.18 | 53,595.93 | 511,887.64 |
| Fresh water | 1,476.86 | 0.00 | 0.00 | 240,536.28 |
| Atmospheric deposition | 994.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 35,982.52 | 9,316.18 | 53,595.93 | 752,423.92 |
| Anticipated crop nutrient removal | 27,974.00 | 4,913.20 | 23,138.90 | 0.00 |
| Actual crop nutrient removal | 22,970.37 | 4,854.93 | 40,097.83 | 207,324.50 |
| Nutrient balance | 13,012.15 | 4,461.26 | 13,498.10 | 545,099.42 |
| Applied to removed ratio | 1.57 | 1.92 | 1.34 | 3.63 |

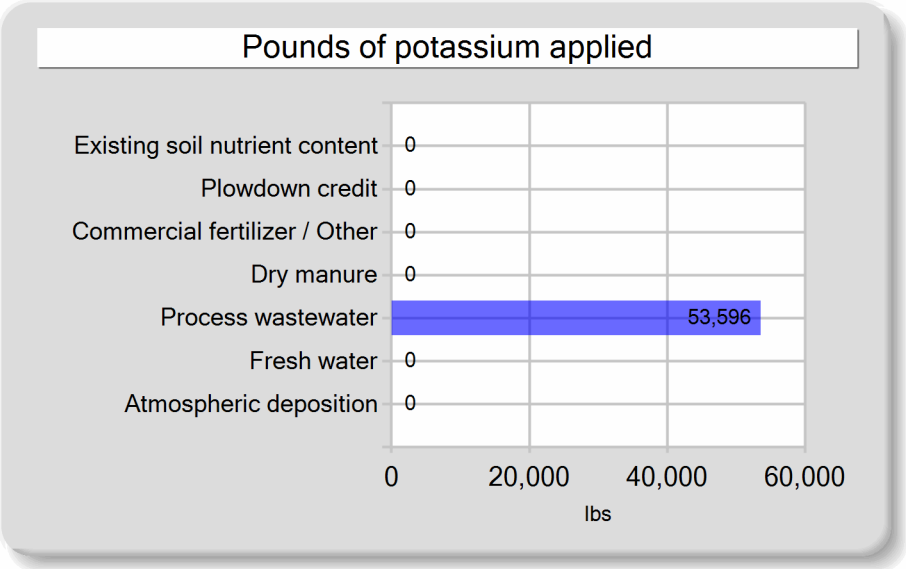
## B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

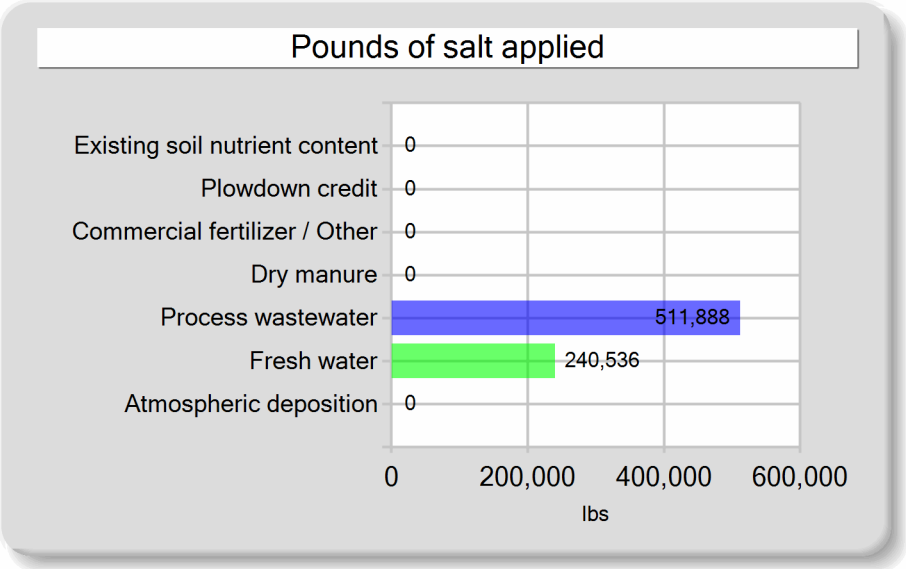


**C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE**



## EXCEPTION REPORTING

### A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

*No manure or process wastewater discharges occurred during the reporting period.*

### B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

*No stormwater discharges occurred during the reporting period.*

### C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

*No land application area to surface water discharges occurred during the reporting period.*

## NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

### A. NUTRIENT MANAGEMENT PLAN STATEMENTS

| | |
|---|---|
| Was the facility's NMP updated in the reporting period? | Yes |
| Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order? | Yes |
| Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order? | Yes |

### B. EXPORT AGREEMENT STATEMENT

| | |
|---|---|
| Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period? | No |

ADDITIONAL NOTES

**A. NOTES**

Field 3 Forage has a nitrogen ratio of 0.00. The crop was not harvested due to field flooding from the San Joaquin River.

**Annual Report - General Order No. R5-2007-0035**
*Reporting period 01/01/2023 to 12/31/2023.*

CERTIFICATION

## A. OWNER AND/OR OPERATOR CERTIFICATION

*I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.*

| | |
|---|---|
| SIGNATURE OF OWNER OF FACILITY | SIGNATURE OF OPERATOR OF FACILITY |
| Ed Machado, Jr. | Edward Anthony Machado |
| PRINT OR TYPE NAME | PRINT OR TYPE NAME |
| 3-5-24 | 3-5-24 |
| DATE | DATE |

**Annual Report - General Order No. R5-2007-0035**
*Reporting period 01/01/2023 to 12/31/2023.*

ATTACHMENTS

## A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

| Annual Dairy Facility Assessment |
| --- |
| Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A ) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed . |

| Manure/Process Wastewater Tracking Manifests |
| --- |
| Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler. |

| Corrective Actions Documents |
| --- |
| Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction. |

| Groundwater Monitoring |
| --- |
| Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13. |

| Storm Water Monitoring |
| --- |
| Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14. |

ATTACHMENT D

## Manure/Process Wastewater Tracking Manifest
## For
## Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination.**
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

**Operator Information:**
Name of Operator: Ed Machado

Name of Dairy Facility: Ed Machado Dairy #2

Facility Address: 5001 Shiloh Road, Modesto, CA 95358

Number and Street | City | Zip Code

Contact Person Name and Phone Number: Ed Machado 209-481-0482

Name | Phone Number

**Manure/Process Wastewater Hauler Information:**
Name of Hauling Company/Person: Rivercrest Cattle

Address of Hauling Company /Person: 11708 E. Moncure Road, Ripon, CA 95366

Number and Street | City | Zip Code

Contact Person: Ed Machado 209-481-0482

Name | Phone Number

**Destination Information:**
Composting Facility / Broker / Farmer / Other (identify) (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

B & M Farms 11708 E. Moncure Road, Ripon, CA 95366 209-599-2565

Name | Number and Street | City | Zip Code | Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

1221 W. Lorenzen Road, Tracy, CA 95304 239-14-008

Number and Street | City | Zip Code | Assessor's Parcel Number

Dates Hauled: 8/15/23 to 10/05/2023

**Amount Hauled:**
Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: 525 Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons): 10.3% moisture

Manure Density (if amount reported in cubic yards):

Method used to determine amount of manure: _____

21 Loads at 25 ton per load

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: _____ Gallons

Method used to determine volume of process wastewater: _____

_____

**Written Agreement:**
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

**Certification:**
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: _____ Date: 3/5/24

Hauler's Signature: _____ Date: 3/5/24

**Denele Analytical, Inc.**

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

# Compliance Analysis Report

**Customer**
Ed Machado Dairy #2
11708 E. Moncure Rd.
Ripon, CA 95366

**Grower**

**Consultant:**
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 11/29/23 12:45 PM
SUBMITTED BY: Vince
ANALYZED DATE: 11/30/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 14.6c

| Sample ID | Lab ID | Analyte | Method Ref | Result | Units |
|---|---|---|---|---|---|
| Dairy Domestic Well | T3333012A | Electrical Conductivity (EC) | EPA 120.1 and | 2.41 | mmhos/cm |
| | | Nitrate Nitrogen (NO3-N) | EPA 353.2 | 60.2 | mg/L |
| | | Ammonia Nitrogen (NH3-N) | SM 4500-NH3 D,C | < 0.500 | mg/L |
| | | Soluble Salts (SALT-SOL) | Calculation | 1540 | mg/L |

| Laboratory Quality Control | LCS % | MS % | MSD % | RPD % |
|---|---|---|---|---|
| Ammonia Nitrogen (NH3-N) | 95.6 | | | |
| Electrical Conductivity (EC) | 96.0 | | | 0.032 |
| Nitrate Nitrogen (NO3-N) | 93.8 | 102 (T3333002A) | 98.3 (T3333002A) | 3.56 |

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.



ELAP Certificate No.: 2714

# DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

## DAIRY COMPLIANCE

Client: Ed Machado Dairy 2
Mailing address: 11708 E. Moncure Rd.
(not facility address) Ripon, CA 95366
Phone#:

Authorized Copy Released To: F+R Ag

Sampled By: Vince Furtado
Project ID:

### ANALYSIS TO BE COMPLETED:

**IRRIGATION/GROUND WATER**
H1 EC*, NO3-N, NH4-N*
H6 EC, TN, TDS
H7 Combination of H1 & H6
H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

**PROCESS WASTE WATER**
H3 EC*, NH4-N, TKN, TP, K, TDS
H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
H8 Combination of H3 &H5
H9 Combination of H4 &H5

**TILE DRAIN**
H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

**DISCHARGE WATER**
E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

**MANURE**
M1 % Moisture
M3 TN TP, K, % Moisture
M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
M7 Combination of M3 & M4

**PLANT TISSUE**
F18 % Moisture, Ash, TN, P, K, DM and as received
F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
F19+ % Moisture, TN (Package includes complete plant tissue analysis)

**SOIL**
K3 Soluble Phosphorus
K4 1ft NO3, Organic Matter
K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
K7 Combination of K3 & K4
K8 Combination of K3 & K5

| | Date Collected | Time Collected | Sample ID | Description | Temp | Well Purge | H1 | H3 | M3 | F18 | OTHER |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 11-29-23 | 9:30 | Dairy Domestic Well | | 14.6°c | | ✓ | | | | |
| 2 | | | | | | | | | | | |
| 3 | | | | | | | | | | | |
| 4 | | | | | | | | | | | |
| 5 | | | | | | | | | | | |
| 6 | | | | | | | | | | | |
| 7 | | | | | | | | | | | |
| 8 | | | | | | | | | | | |
| 9 | | | | | | | | | | | |
| 10 | | | | | | | | | | | |

Analysis

### CHAIN OF CUSTODY RECORDING

| | Signature | Company | Received Date & Time | Relinquished Date & Time | (LABORATORY USE ONLY) Logged In By: | LABORATORY NUMBER |
|---|---|---|---|---|---|---|
| 1st | Vince Furtado | F+R Ag | | 11-29-23 | | |
| 2nd | | DENELE ANALYTICAL, INC. | 11-29-23 12:45 | | | T333012 |

Required to be submitted to any agency? ( YES / NO )